

Agri_Empower Training Series

Yam Mastery: Advanced Level (Industrial Scale & Ag-Tech)

Module 1: Production & Management

Precision Mechanization & Irrigation

- **Mechanized Ridge Formation:** Utilizing tractor-mounted ridgers for standardized 75cm-100cm mounds, ensuring uniform drainage across 10+ hectares.
- **Dry-Season Production:** Deployment of Drip Irrigation systems. Focus on water budgeting: delivering exactly 4-5 liters per plant daily during peak heat.
- **Industrial Trellis Systems:** Moving from individual stakes to wire-and-pole trellising. This mimics a vineyard, allowing for better airflow and 30% more sunlight penetration.

Module 2: Pest, Disease & Soil Health

Data-Driven Protection

- **Foliar Nutrition:** Direct leaf application of Liquid NPK + Micronutrients (Boron/ Zinc) to boost tuber bulking in the 4th month.
- **Bio-Nematocides:** Using industrial-grade Neem extracts and Marigold crop rotation to eliminate "Tuber Cracking" caused by nematodes.

- **GPS Soil Mapping:** Creating a digital fertility map to apply fertilizer only where needed, reducing chemical waste by 20%.

Module 3: Post-Harvest & Storage

The Cold Chain Strategy

- **Evaporative Cooling Systems:** Maintaining storage at 15°C with 85% humidity to prevent weight loss (shrinkage) over 6-9 months.
- **Phytosanitary Curing:** Using automated curing chambers to heal harvest wounds, essential for meeting international phytosanitary export standards.
- **Palletization:** Standardizing tuber crates for forklift handling, minimizing the human touch to reduce bruising.

Module 4: Processing & Value Addition

Industrial Grade Standards

- **Starch Extraction:** Industrial centrifugation techniques to produce high-purity yam starch for pharmaceuticals.
- **Flash Drying:** Replacing sun-drying with high-speed automated dryers to produce "Grade A" white flour (Poundo Yam) with zero microbial count.
- **Sustainable Waste Management:** Processing peels into high-protein animal feed pellets using pelletizing machines.

Module 5: Agribusiness & Markets

Supply Chain Excellence

- **Export Compliance (Global G.A.P.):** Documentation of every chemical spray, worker safety protocols, and water quality tests for the EU/US markets.
- **Contract Farming:** Negotiating forward contracts with supermarkets to fix prices before the crop is even planted.

Advanced Competency Test: Trainees must design a 1-hectare irrigation schedule and prepare a "Technical Farm Records" folder for a bank audit.